

Gauge Theory in $\mathcal{UD}$

Brian Beckman

Feb 2026

Abstract

This notebook extends the Relativity Toolkit (v1.11.0) to basic Abelian scalar gauge theory. We implement a generalized covariant derivative operator that simultaneously processes spacetime geometry and internal phase geometry. We demonstrate this implementation through four automated computational proofs:

- *Metric Compatibility*: the Levi-Civita connection uniquely nulls the gradient of the metric ($\nabla g = 0$)
- *$U(1)$ Gauge Covariance*: the covariant derivative of a scalar field transforms covariantly under local phase rotation.
- *Field Strength Invariance*: the electromagnetic tensor $F_{\mu\nu}$ is invariant under local gauge transformations.
- *Charged Vector Field Covariance*: the covariance of a generic charged vector field V^α transforms covariantly under simultaneous spacetime coordinate changes and internal $U(1)$ phase rotations.

Finally, we apply the updated engine to re-derive the Riemann curvature tensor from the torsion-free commutator of covariant derivatives.

Initialize the Relativity Toolkit

Optional: Quit for a Fresh Kernel

- *Note, if you call `Quit` in a notebook, evaluation may hang. If you want to “Evaluate Notebook,” mark the following two cells “Evaluatable” in the Cell→Property menu, call `Quit` and `FrontEndTokenExecute` once each, then comment out the cells, or mark them unevaluatable again. Alternatively, you can quit the kernel via the Evaluation menu (last item). At that point, you can “Evaluate Notebook” in a clean environment, or evaluate cells individually.*

```
Quit[];
```

```
In[ ]:= FrontEndTokenExecute["DeleteGeneratedCells"]
```

Download the Code

```
In[1]:= RTbranch = "v1.11.0";
RTbase = "https://raw.githubusercontent.com/rebcabin/RelativityToolkit/" <>
RTbranch <> "/";
RTbust = "?t=" <> ToString[Round[AbsoluteTime[]]]; (*Force fresh download*)
RTrules = RTbase <> "RelativityToolkit.wl" <> RTbust;
RTtests = RTbase <> "RelativityToolkit_RegressionTests.wl" <> RTbust;
RTviz = RTbase <> "RelativityToolkit_VisualShowcase.wl" <> RTbust;
Get[RTrules];

» Relativity Toolkit version : 1.11.0
» Relativity Connection Symbol :  $\Gamma$ 
```

Optional: See all Tests and Results from Prior Works

Make any of the following cells evaluable if you wish. They're unevaluable by default to keep this notebook trim.

```
In[*]:= Get[RTtests];

In[*]:= Get[RTviz] (* no semicolon, on purpose *)

In[*]:= Get[RTbase <> "covariant_derivative.wl"]

In[*]:= Get[RTbase <> "law_of_gamma.wl"]

In[*]:= Get[RTbase <> "connections_to_christoffels.wl"]

Get[RTbase <> "schwarzschild_from_scratch.wl"]
```

Introduction

In physics, choosing a coordinate system or a phase reference is “setting a gauge.” Changing this choice—whether transforming spacetime coordinates or shifting the phase of a field—often introduces inhomogeneous terms when taking partial derivatives.

Connection fields arise to maintain covariance of the physics under such transformations. In general relativity, the Levi-Civita connection (Christoffel symbols) corrects for the curvature of spacetime coordinates. In electrodynamics, the vector potential A^μ acts as the connection, correcting for local variations in the phase reference.

While apparent forces (like centrifugal and Coriolis) can be eliminated globally by choosing a specific inertial gauge, fundamental forces (like tidal gravity and electromagnetism) arise from the intrinsic curvature of these connections and cannot be globally gauged away.

Covariance and Internal Gauge

To see how a connection restores covariance, consider a scalar matter field ψ subjected to a local $U(1)$ phase change $\Lambda(x^\mu)$:

$$\bar{\psi} = e^{iq\Lambda} \psi \quad (1)$$

where q is a constant charge. The ordinary partial derivative produces an inhomogeneous term via the product and rules:

$$\partial_\mu \bar{\psi} = e^{iq\Lambda} (\partial_\mu \psi + i q (\partial_\mu \Lambda) \psi) \quad (2)$$

The inhomogeneous term $i q (\partial_\mu \Lambda) \psi$ breaks covariance. To restore it, introduce a gauge covariant derivative. As with general relativity, add a correction to the partial derivative

$$D_\mu \psi \equiv \partial_\mu \psi - i q A_\mu \psi \quad (3)$$

Requiring the connection to transform as

$$\bar{A}_\mu = A_\mu + \partial_\mu \Lambda \quad (4)$$

gives us an operator D_μ that transforms covariantly with gauge, just as ∇_μ transforms covariantly with coordinates in relativity:

$$\begin{aligned} D_\mu \bar{\psi} &= \partial_\mu \bar{\psi} - i q \bar{A}_\mu \bar{\psi} = \\ &= e^{iq\Lambda} (\partial_\mu \psi + i q (\partial_\mu \Lambda) \psi) - i q (A_\mu + \partial_\mu \Lambda) e^{iq\Lambda} \psi = \\ &= e^{iq\Lambda} (\partial_\mu \psi - i q A_\mu \psi) = \\ &= e^{iq\Lambda} D_\mu \psi \end{aligned} \quad (5)$$

Internal Gauge Spaces in the $\mathcal{UD}$ Calculus

In general relativity, we take covariant derivatives of tensors inhabiting the tangent spaces of a curved manifold. The connection Γ accounts for the twisting of basis vectors.

In gauge theories like electrodynamics, the connections operate on internal spaces (such as the $U(1)$ phase) attached, extrinsically to the spacetime itself, to each point in spacetime. The objective of this notebook is to update the $\mathcal{UD}$ calculus to compute derivatives for objects that possess both spacetime indices and internal gauge charges simultaneously.

Covariant Derivatives, Revisited

In GR, the canonical example is the covariant derivative of an arbitrary contravariant vector A^μ with respect geometrical coordinates:

```
In[8]:= With[{example = HoldForm[CD[A[U[m]], x[U[v]]]]},
  Echo[TensorForm@@example, example ""];
» (Aμ;v) Aμ;v + (Aλ) Γμvλ
```

The connection coefficients Γ account for twisting and turning of basis vectors in the tangent spaces as the vector A^μ is parallel-transported geometrically. In GR, this twisting and turning causes geodesics to converge or diverge, thus causing tidal forces. Twisting and turning also drives single geodesics, giving

rise to orbits, gravitational lensing, gravitational waves, and so on.

- *Emphasis: Geodesic deviation is indistinguishable from tidal force. It's not an apparent force: its numerical values may depend on gauge choice, but we can't get rid of it globally by gauge choice.*

The derivative above transforms like a tensor, ensuring physics is **locally Lorentz invariant**. This invariance invites Noether's theorem again, implying a conserved quantity. In GR, this conserved quantity is **spin current**, from symmetries of non-coordinate bases or **tetrads** (see the Appendix).

A New CD

The following new overload of CD computes covariant derivatives for Abelian (commutative) gauges like scalar phase. We'll get to non-Abelian gauges like $SO(3)$ and $SU(2)$ in future notebooks.

Gauge Covariant Derivative

Inputs

1. **expr**: The field being differentiated, scalar or tensor
2. **var**: the coordinate function, e.g., $x[\mathcal{U}[\mu]]$, the differentiation variable
3. **q**: the coupling constant or "charge"
4. **A**: the gauge-field connection symbol. Γ is kept implicitly as the spacetime connection for all theories.

```
In[9]:= CD[expr_, var_, q_, A_] :=
Module[{gravitationalPart, gaugeCorrection, idx},
  (*Get coordinate covariant derivative via existing CD overloads.*)
  (* $\partial$  + Christoffels (if expr is a tensor)*)
  (*If expr is a scalar, just return  $\partial$ .**)
  gravitationalPart = CD[expr, var];
  (*Extract the index from the differentiation variable.*)
  (*var is some  $x[\mathcal{U}[\mu]]$ ; we need  $\mu$ *)
  idx = var[[1, 1]];
  (*Construct the gauge correction, e.g.,  $-i q A_\mu * \text{expr}$ *)
  (*Note: This A has a DOWN (Covariant) index matching the derivative*)
  gaugeCorrection = -I * q * A[D[idx]] * expr;
  (*Add*)
  gravitationalPart + gaugeCorrection];
```

Electrodynamics Demos

Covariance of the Covariant Derivative

Above, we subjected ψ to gauge transformation manually. Automat that calculation:

```
In[10]:= (cde = CD[ψ, x[ℳ[μ]], q, A])
```

```
Out[10]= -i q ψ (Aμ) + ψ,μ
```

Starting with the gauge transformations

$$\bar{\psi} = e^{iq\Lambda} \psi \quad \text{and} \quad \bar{A}_\mu = A_\mu + \partial_\mu \Lambda$$

```
In[11]:= gaugeTransforms = {
  ψ → ei q Λ ψ,
  A[ℳ[μ_]] :=> A[ℳ[μ]] + Partials[Λ, x[ℳ[μ]]]}
```

```
Out[11]= {ψ → ei q Λ ψ, Aμ_ :=> Aμ + Λ,μ}
```

prove that the covariant derivative

$$D_\mu \psi \equiv \partial_\mu \psi - iq A_\mu \psi$$

transforms exactly as does ψ

$$D_\mu \bar{\psi} = e^{iq\Lambda} D_\mu \psi$$

Need one rule for the charge constant. Everything else is already in the toolkit's

differentiationRules:

```
In[12]:= constantRules = {Partials[q, x[ℳ[μ_]]] :=> 0}
```

```
Out[12]= {q,μ_ :=> 0}
```

```
In[13]:= transformedCde =
  (cde /. gaugeTransforms /. differentiationRules /. constantRules) // Expand //
  Simplify
```

```
Out[13]= ei q Λ (-i q ψ (Aμ) + ψ,μ)
```

```
In[14]:= transformedCde === ei q Λ cde
```

```
Out[14]= True
```

QED ■

Invariance of the Field-Strength Tensor

The field-strength tensor, $F_{\mu\nu}$, aka **Faraday**, is defined as follows

```
In[15]:= F[μ, ν] = Partials[A[ℳ[ν]], x[ℳ[μ]]] - Partials[A[ℳ[μ]], x[ℳ[ν]]]
```

```
Out[15]= -(Aμ,ν) + Aν,μ
```

usually presented as $\partial_\mu A_\nu - \partial_\nu A_\mu$ to make the indices line up.

Now, subject it to the gauge transform:

```
In[16]:= F[μ, ν] /. gaugeTransforms
```

```
Out[16]= - (Aμ + Δ,μ),ν + (Aν + Δ,ν),μ
```

See that it doesn't change by invoking the toolkit's `ExpandDerivatives`:

```
In[17]:= (F[μ, ν] /. gaugeTransforms) // ExpandDerivatives
```

```
Out[17]= - (Aμ,ν) + Aν,μ
```

QED ■

Charged Vector Field Covariance Demo

In this demo, we differentiate a generic charged contravariant vector field V^α coupled to the electromagnetic gauge field A_μ by a scalar charge q .

Because V^α is a hybrid mathematical object—possessing both internal phase and a geometrical space-time index—its covariant derivative requires two distinct corrections. A_μ acts as the connection for the internal phase, while the Christoffel symbols Γ act as the connection for the curved spacetime geometry.

We did not need a geometrical correction in the previous electrodynamics demo because we were differentiating a scalar matter field ψ , which has no geometric indices. Here, the vector V^α suffers both gauge transformation and geometrical coordinate transformation.

Proceed just as above:

```
In[18]:= (vsd = CD[V[U[α]], x[U[μ]], q, A]) // TensorForm
```

```
Out[18]= Vα,μ - i q (Aμ) (Vα) + (Vλ) Γαμλ
```

```
In[19]:= mixedGaugeTransforms = {
  V[U[α_]] -> ei q Δ * V[U[α]],
  A[D[μ_]] -> A[D[μ]] + Partial[Δ, x[U[μ]]]}
```

```
Out[19]= {Vα -> ei q Δ (Vα), Aμ -> Aμ + Δ,μ}
```

```
In[20]:= transformedVsd =
  (vsd /. mixedGaugeTransforms /. differentiationRules /. {Partial[q, _] -> 0}) //
  Expand // TensorForm
```

```
Out[20]= ei q Δ (Vα,μ) - i ei q Δ q (Aμ) (Vα) + ei q Δ (Vλ) Γαμλ
```

```
In[21]:= expectedVsd = vsd * ei q Δ // Expand // TensorForm
```

```
Out[21]= ei q Δ (Vα,μ) - i ei q Δ q (Aμ) (Vα) + ei q Δ (Vλ) Γαμλ
```

```
In[22]:= transformedVsd === expectedVsd
Out[22]= True
```

QED ■

Geometrical GR Demos

Geometrical Covariant Derivative of a Scalar Function Λ

equal to its ordinary partial derivative, rendered in comma notation

```
In[23]:= CD[Λ, x[U[μ]]]
Out[23]= Λ,μ
```

Geometrical Covariant Derivative of a Contravariant Vector V^μ

equal to its partial derivative plus the correction proportional to the geometrical connection Γ

```
In[24]:= CD[V[U[μ]], x[U[v]]] // TensorForm
Out[24]= V^μ,ν + (V^λ) Γ^μ_νλ
```

Geometrical Covariant Derivative of a (2, 0) Tensor

now we get two corrections, one for each index

```
In[25]:= CD[T[U[μ], U[v]], x[U[ρ]]] // TensorForm
Out[25]= T^μν,ρ + (T^λν) Γ^μ_ρλ + (T^μλ) Γ^ν_ρλ
```

Geometrical Covariant Derivative of a (1, 1) Tensor

ditto, with a sign change

```
In[26]:= CD[T[U[μ], D[v]], x[U[ρ]]] // TensorForm
Out[26]= T^μν,ρ - (T^μλ) Γ^λ_νρ + (T^λν) Γ^μ_ρλ
```

Geometrical Covariant Derivative of the Covariant Metric

Two negative signs, one for each covariant index

```
In[27]:= (cdg = CD[g[D[μ], D[v]], x[U[ρ]]]) // TensorForm
Out[27]= g_μν,ρ - (g_λν) Γ^λ_μρ - (g_μλ) Γ^λ_νρ
```

Metric Compatibility, $\nabla g = 0$

$\nabla g = 0$ was an axiom in earlier notebooks, and it yielded the Christoffel symbols in the *Fundamental Theorem of Riemannian Geometry*. Here, we do the opposite, showing that the Christoffel symbols imply metric compatibility. Thus, in a torsion-free manifold, metric compatibility is both necessary and sufficient for the Christoffel symbols.

- In abstract geometrical contexts, as opposed to coordinate-basis contexts, the Christoffel symbols are called the **Levi-Civita Connection**., often written $\left\{ \begin{smallmatrix} \lambda \\ \mu \nu \end{smallmatrix} \right\}$.
- Christoffel symbols were derived in Covariant Derivatives and Connections in the $\mathcal{UD}$ Calculus, assuming both torsion freedom (symmetry in lower two indices of Γ), and metric compatibility.
- If we allow torsion, as in Einstein-Cartan theory or as in the Appendix, the present tautology is not valid. In such cases, we must introduce the **contortion tensor** K

$$\Gamma^\lambda_{\mu\nu} = \left\{ \begin{smallmatrix} \lambda \\ \mu \nu \end{smallmatrix} \right\} + K^\lambda_{\mu\nu} \quad (6)$$

We do not develop this case further in this notebook.

For pedagogy, clear the definitions loaded from the toolkit:

```
In[28]:= ClearAll[ruleLeviCivita, cdgWithLeviCivitaConnection, metricDifferentiationRules];
```

Start with the covariant derivative of the metric, `cdg`, from above

Fetch the Levi-Civita connection via a new `ruleLeviCivita`, added to Relativity Toolkit v1.9.0:

$$\Gamma^\lambda_{\mu\rho} = \frac{1}{2} g^{\lambda\sigma} (g_{\sigma\rho,\mu} + g_{\sigma\mu,\rho} - g_{\mu\rho,\sigma}) \quad (7)$$

```
In[29]:= ruleLeviCivita = r[u[lam_], d[a_], d[b_]] =>
```

```
Module[{sigma = Unique["sigma"]},
  1/2 g[u[lam], u[sigma]] *
  (Partials[g[d[sigma], d[b]], x[u[a]]] +
   Partials[g[d[sigma], d[a]], x[u[b]]] -
   Partials[g[d[a], d[b]], x[u[sigma]])]
```

```
Out[29]=
```

$$\Gamma^\lambda_{a_b} \Rightarrow \text{Module}\left[\{\sigma = \text{Unique}[\sigma]\}, \frac{1}{2} (g^{\lambda \sigma 23}) (g_{\sigma 23 \ b,a} + g_{\sigma 23 \ a,b} - g_{a \ b, \sigma 23})\right]$$

In `cdg`, replace Γ with its definition via this rule, then `Expand` to expose terms to further processing


```
In[30]:= (cdgWithLeviCivitaConnection = (cdg /. ruleLeviCivita) // Expand) // TensorForm
```

```
Out[30]=
```

$$\begin{aligned}
& -\frac{1}{2} (g_{\lambda \nu}) (g^{\lambda \kappa}) (g_{\kappa \mu, \rho}) - \frac{1}{2} (g_{\mu \lambda}) (g^{\lambda \kappa}) (g_{\kappa \nu, \rho}) - \\
& \frac{1}{2} (g_{\lambda \nu}) (g^{\lambda \kappa}) (g_{\kappa \rho, \mu}) - \frac{1}{2} (g_{\mu \lambda}) (g^{\lambda \kappa}) (g_{\kappa \rho, \nu}) + \\
& g_{\mu \nu, \rho} + \frac{1}{2} (g_{\lambda \nu}) (g^{\lambda \kappa}) (g_{\mu \rho, \kappa}) + \frac{1}{2} (g_{\mu \lambda}) (g^{\lambda \kappa}) (g_{\nu \rho, \kappa})
\end{aligned}$$

Now consider the following new `metricDifferentiationRules` in the toolkit:

They include

- metric symmetry: $g_{\mu\nu} = g_{\nu\mu}$, required by standard general relativity
- metric contraction to the Kronecker delta: $g_{\lambda\nu} g^{\lambda\mu} = \delta_{\nu}^{\mu}$ ($gUpUp$ is inverse of $gDownDown$)
- contraction of Kronecker delta on derivatives, for the chain rule: $\delta_{\nu}^{\mu} \partial_{\mu} = \partial_{\nu}$
- delta contraction inside metric derivatives: $\delta_{\nu}^{\mu} g_{\mu\lambda, \rho} = g_{\nu\lambda, \rho}$, $\delta_{\nu}^{\mu} g_{\lambda\mu, \rho} = g_{\lambda\nu, \rho}$

```
In[31]:= metricDifferentiationRules = {
  (*Metric Symmetry*)
  g[D[a_], D[b_]] /; ! OrderedQ[{a, b}] >=> g[D[b], D[a]],
  g[U[a_], U[b_]] /; ! OrderedQ[{a, b}] >=> g[U[b], U[a]],
  (*Permissive Contraction (All 4 Alignments)*)
  (*First-First*)
  g[D[s_], D[a_]] * g[U[s_], U[b_]] >=> delta[U[b], D[a]],
  (*Second-Second*)
  g[D[a_], D[s_]] * g[U[b_], U[s_]] >=> delta[U[b], D[a]],
  (*First-Second (Redundant but Safe)*)
  g[D[s_], D[a_]] * g[U[b_], U[s_]] >=> delta[U[b], D[a]],
  (*Second-First*)
  g[D[a_], D[s_]] * g[U[s_], U[b_]] >=> delta[U[b], D[a]],
  (*Delta Contractions*)
  delta[U[bound_], D[free_]] * Partials[expr_, x[U[bound_]]] >=>
    Partials[expr, x[U[free]]],
  delta[U[bound_], D[free_]] * Partials[g[D[bound_], D[other_]], v_] >=>
    Partials[g[D[free], D[other_]], v],
  delta[U[bound_], D[free_]] * Partials[g[D[other_], D[bound_]], v_] >=>
    Partials[g[D[other], D[free]], v]};
```

The result $\nabla g = 0$ is immediate upon application of both the old metric rules and the new metric rules:

```
In[32]:= 0 ==
```

```
(cdgWithLeviCivitaConnection /. (metricRules ~ Join ~ metricDifferentiationRules))
```

```
Out[32]=
```

True

QED ■

Riemann Revisited

Here is a concise derivation of the Fundamental Theorem of Riemannian Geometry, shorter than the derivation in our earlier notebooks

- *Because we are using the symmetric Levi-Civita connection implicitly via `torsionRules`, the commutator reduces to Riemann curvature tensor contracted with A . If we allowed torsion, as discussed in the Appendix, an extra term proportional to the torsion tensor would appear here.*

Calculate the commutator: $[\nabla_\rho, \nabla_\nu] A^\mu$:

```
In[33]:= term1 = CD[CD[A[u[mu]], x[u[nu]]], x[u[rho]]];
term2 = CD[CD[A[u[mu]], x[u[rho]]], x[u[nu]]];
(comm = term1 - term2) // TensorForm

Out[35]=
```

$$\left(\left(A^\lambda \right) \Gamma_{\nu\lambda}^\mu \right)_{,\rho} - \left(\left(A^\lambda \right) \Gamma_{\rho\lambda}^\mu \right)_{,\nu} - \left(A^\mu \right)_{,\lambda} \Gamma_{\nu\rho}^\lambda + \left(A^\mu \right)_{,\lambda} \Gamma_{\rho\nu}^\lambda - \left(A^\lambda \right) \Gamma_{\nu\rho}^\kappa \Gamma_{\kappa\lambda}^\mu +$$

$$\left(A^\lambda \right) \Gamma_{\rho\nu}^\kappa \Gamma_{\kappa\lambda}^\mu - \left(A^\lambda \right) \Gamma_{\rho\lambda}^\kappa \Gamma_{\nu\kappa}^\mu - \left(A^\lambda \right)_{,\rho} \Gamma_{\nu\lambda}^\mu + \left(A^\lambda \right) \Gamma_{\nu\lambda}^\kappa \Gamma_{\rho\kappa}^\mu + \left(A^\lambda \right)_{,\nu} \Gamma_{\rho\lambda}^\mu$$

Simplification Pipeline

```
In[36]:= calculatedRiemann =
  ((comm //
    ExpandDerivatives //
    Expand //
    CanonicalizeIndices) //.
    torsionRules) //
  ExtractCoefficient[#, A] & //
  TensorForm

Out[36]=
```

$$\Gamma_{\lambda\nu,\rho}^\mu - \Gamma_{\lambda\rho,\nu}^\mu - \Gamma_{\lambda\rho}^\kappa \Gamma_{\kappa\nu}^\mu + \Gamma_{\lambda\nu}^\kappa \Gamma_{\kappa\rho}^\mu$$

Textbook definition: $\Gamma_{\nu\check{s},\rho}^\mu - \Gamma_{\rho\check{s},\nu}^\mu + \Gamma_{\rho\lambda}^\mu \Gamma_{\nu\check{s}}^\lambda - \Gamma_{\nu\lambda}^\mu \Gamma_{\rho\check{s}}^\lambda$

```
In[37]:= rawTextbook = Partials[r[u[mu], d[nu], d[s]], x[u[rho]]] -
  Partials[r[u[mu], d[rho], d[s]], x[u[nu]]] +
  r[u[mu], d[rho], d[lambda]] * r[u[lambda], d[nu], d[s]] -
  r[u[mu], d[nu], d[lambda]] * r[u[lambda], d[rho], d[s]]

Out[37]=
```

$$\Gamma_{\nu\check{s},\rho}^\mu - \Gamma_{\rho\check{s},\nu}^\mu - \Gamma_{\rho\check{s}}^\lambda \Gamma_{\nu\lambda}^\mu + \Gamma_{\nu\check{s}}^\lambda \Gamma_{\rho\lambda}^\mu$$

```
In[38]:= expectedRiemann = (rawTextbook //. torsionRules) // TensorForm

Out[38]=
```

$$\Gamma_{\lambda\nu,\rho}^\mu - \Gamma_{\lambda\rho,\nu}^\mu - \Gamma_{\lambda\rho}^\kappa \Gamma_{\kappa\nu}^\mu + \Gamma_{\lambda\nu}^\kappa \Gamma_{\rho\kappa}^\mu$$

```
In[39]:= calculatedRiemann === expectedRiemann

Out[39]=
```

True

QED ■

Appendix: Spin Connection as Gravitational Gauge

General relativity is often compared to standard gauge theories, but there is a fundamental distinction. In theories like electromagnetism or quantum chromodynamics, forces arise from connections on internal spaces (e.g., $U(1)$ or $SU(3)$) defined over spacetime. In standard general relativity, gravitational forces arise from the Levi-Civita connection, which is inextricably linked to the tangent bundle of the spacetime geometry itself.

To formulate gravitation as a strict gauge theory functionally equivalent to the Standard Model forces, one must identify the conserved currents corresponding to local gauge symmetries. The definitive literature demonstrates how this is achieved:

- Friedrich W. Hehl et al., *General Relativity with Spin and Torsion: Foundations and Prospects* (Rev. Mod. Phys. 48, 393, 1976)

which shows that:

- Invariance under general transformations (diffeomorphisms), implies conservation of the energy-momentum tensor.
- Invariance under local Lorentz transformations implies conservation of the spin angular momentum tensor (the spin current).
- Standard Einstein GR with vanishing torsion, which we assumed in this series of notebooks, collapses the two symmetries and hides the independent spin current.

Some other references are

- Sean Carroll, *Spacetime and Geometry: An Introduction to General Relativity*

- Appendix J, Page 486ff, under local Lorentz invariance, tetrads, and the spin connection:

Spin connection ω_μ^{ab} , arising from non-coordinate basis sets, acts as the gauge field for local Lorentz transformations. Carroll also explains that in standard GR the torsion tensor $T^\lambda_{\mu\nu}$ is zero, locking the Levi-Civita connection (Christoffel symbols) to the metric and forcing the energy-momentum tensor to be symmetric, absorbing spin effects via symmetrization.

- Also see Hehl above, Equation 3.8 and Sections III.B and III.C. In other references, this may be called *Belinfante-Rosenfeld symmetrization*.

- T. W. B. Kibble, *Lorentz Invariance and the Gravitational Field* (J. Math. Phys. 2, 212, 1961)

- *Foundational paper deriving gravitation as the gauge theory of the Poincaré group, i.e., translations + Lorentz*

Kibble shows that (1) gauging *translations* in Poincaré yields the tetrad field, defining the metric and conserving energy-momentum, and that (2) gauging the Lorentz *boost+rotation subgroup* yields the spin connection, permitting transformation of spinors, otherwise impossible.

Conclusion

The Relativity Toolkit (v1.11.0) can process hybrid objects, such as the charged vector field V^α , with simultaneous geometric and internal phase corrections. While this notebook focuses on the Abelian

$U(1)$ gauge group, the underlying computational logic is a foundation for future non-Abelian gauge theories.

References

1. Beckman, Brian. *From the steam age to quantum fields: a unified approach via UD tensorial calculus*. Wolfram Community post.
2. Beckman, Brian. *General relativity in the UD calculus*. Wolfram Community post.
3. Beckman, Brian. *Formal Differential Geometry in the UD Calculus: Part 1*. Wolfram Community post.
4. Beckman, Brian. *Covariant Derivatives and Connections in the UD Calculus*. Wolfram Community post.
5. Beckman, Brian. *Riemann Curvature in UD + Compiler POC*. Wolfram Community post.
6. Beckman, Brian. *Schwarzschild Orbits in the UD Calculus*. Wolfram Community post.
7. Beckman, Brian. *Schwarzschild Metric from Scratch via UD Compilation*. Wolfram Community post (to appear).
8. Wolfram Function Repository. *MetricTensor*. For component-based calculations.
9. Wolfram Function Repository. *ChristoffelSymbol*. For component-based calculations.
10. [MTW] Misner, Charles W.; Thorne, Kip S.; Wheeler, John Archibald. *Gravitation*. Princeton University Press, 2017.
11. Sean Carroll, *Spacetime and Geometry: An Introduction to General Relativity*. Cambridge University Press, 2019.
12. Friedrich W. Hehl et al., *General Relativity with Spin and Torsion: Foundations and Prospects* (Rev. Mod. Phys. 48, 393, 1976)
13. T. W. B. Kibble, *Lorentz Invariance and the Gravitational Field* (J. Math. Phys. 2, 212, 1961)
14. Sciama, D. W., *On the analogy between charge and spin in general relativity*. In: Recent Developments in General Relativity, p. 415. Pergamon, New York, 1962.